



MANGO



Visvesvaraya Trade Promotion Centre (VTPC)

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Preface

With a robust agro potential the Country has, the Government of India has launched the One District One Focus Product Scheme (ODOFP) for agriculture sector by the Ministry of Food Processing Industries. Among its primary objectives, the ODOFP initiative aims to enhance the value of the products which is eventually expected to boost employment and income levels of the farmers. The identified products across the country is available at <https://www.nfsm.gov.in/odopstatecropsreport.aspx>

Karnataka has identified its product mix with high demand and export potential across all the districts. The implementing agency for this initiative is the Karnataka State Agricultural Produce Processing and Export Corporation Limited (KAPPEC), the nodal agency in the State.

VTPC being the nodal agency for promotion of exports from the State has made an attempt to draw the action plan district wise, to capture the potential, present status and future prospects in domestic and international markets. Besides, detailed insights have been provided into the biological description of the product, their local, national and international varieties, export-import analysis, HS codes. For a holistic perspective for those concerned, each report also provides SPS standards, processing technologies available, export grading and packing specifications, and suggested pivotal roles and responsibilities among the government departments, boards, corporations and Universities.

Presenting the Action Plan/Report for Mango, a ODOFP product mapped to Haveri and Dharwad districts of Karnataka, formulated by VTPC. I sincerely hope that this ready reckoner with first-hand information regarding the farming sector would help all those concerned, especially the FPOs and food processing entrepreneurs who have a desire to focus on exporting this product.

S.R. Satheesha

Director (Exports) & Managing Director

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Introduction

Mango is an edible stone fruit produced by the tropical tree *Mangifera indica* which is believed to have originated in the region between northwestern Myanmar, Bangladesh, and northeastern India. *M. indica* has been cultivated in South and Southeast Asia since ancient times resulting in two types of modern mango cultivars: the "Indian type" and the "Southeast Asian type". Other species in the genus *Mangifera* also produce edible fruits that are also called "mangoes", the majority of which are found in the Malesian ecoregion. Mango is the national fruit of India, Pakistan and the Philippines, while the mango tree is the national tree of Bangladesh.

Physical description

Mango trees grow to 30–40 meters (98–131 feet) tall, with a crown radius of 10–15 m (33–49 feet). The trees are long-lived, as some specimens still fruit after 300 years.

In deep soil, the taproot descends to a depth of 6 m (20 ft), with profuse, wide-spreading feeder roots and anchor roots penetrating deeply into the soil. The leaves are evergreen, alternate, simple, 15–35 cm (6–14 inches) long, and 6–16 cm broad; Young leaves are orangish-pink, rapidly changing to a dark, glossy red, then dark green as they mature. The flowers are produced in terminal panicles which are 10–40 cm in length; Individual flower is small and white with five petals of 5–10 mm in length, with a mild, sweet fragrance. Over 500 varieties of mangoes are known, many of which ripen in summer, while some give a double crop. The fruit takes four to five months from flowering to ripen naturally.

The ripe fruit varies according to cultivar in size, shape, colour, sweetness, and eating quality. Depending on the cultivar, fruits are variously yellow, orange, red, or green. The fruit has a single flat, oblong pit that can be fibrous or hairy on the surface and does not separate easily from the pulp. The fruits may be somewhat round, oval, or kidney-shaped, ranging from 5–25 cms (2–10 in) in length and from 140 grams (5 oz) to 2 kilograms (5 lb) in weight per individual fruit. The skin is leather-like, waxy, smooth, and fragrant, with colours ranging from green to yellow, yellow-orange, yellow-red, or blushed with various shades of red, purple, pink or yellow when fully ripe.

Ripe intact mangoes give off a distinctive resinous, sweet smell. Inside the pit is a thin lining of 1–2 mm (0.039–0.079 in) thickness, covering a single seed which is 4–7 cm (1.6–2.8 in) in length. Mangoes have recalcitrant seeds which do not survive freezing and drying. Mango trees grow readily from seeds, with germination success highest when seeds are obtained from mature fruits.



Phytochemistry

Mango is an edible stone fruit produced by the tropical tree *Mangifera indica* which is believed to have originated in the region between northwestern Myanmar, Bangladesh, and northeastern India. *M. indica* has been cultivated in South and Southeast Asia since ancient times resulting in two types of modern mango cultivars: the "Indian type" and the "Southeast Asian type". Other species in the genus *Mangifera* also produce edible fruits that are also called "mangoes", the majority of which are found in the Malesian ecoregion. Mango is the national fruit of India, Pakistan and the Philippines, while the mango tree is the national tree of Bangladesh.

Nutrition

Mango is an edible stone fruit produced by the tropical tree *Mangifera indica* which is believed to have originated in the region between northwestern Myanmar, Bangladesh, and northeastern India. *M. indica* has been cultivated in South and Southeast Asia since ancient times resulting in two types of modern mango cultivars: the "Indian type" and the "Southeast Asian type". Other species in the genus *Mangifera* also produce edible fruits that are also called "mangoes", the majority of which are found in the Malesian ecoregion. Mango is the national fruit of India, Pakistan and the Philippines, while the mango tree is the national tree of Bangladesh.



Popular Mango varieties across the world

Worldwide, there are several hundred cultivars of mango. Depending on the cultivar, mango fruit varies in size, shape, sweetness, skin color, and flesh color which may be pale yellow, gold, green, or orange.



1.Honey (Ataulfo)

Honey mangoes have a combination of sweet and sour flavor with smooth and firm flesh without fibers. Fruit pulp has a peachy tropical aroma, and it encloses a small pit. Honey mangoes are available from March to July. The primary sources are Mexico, Brazil, Peru, and Ecuador. When fully ripe, the fruit turns deep yellow with small wrinkles and acquires a flattened oblong shape.

2.Francis

These mangoes have sweet and fruity flavour. Flesh of the fruit is juicy, soft, and fibrous and has a peach-like tropical fragrance. One can have this luscious mango from May to June. The primary sources are Haiti and Ecuador. The fruit grows with green overtones that become yellow and change to more golden when mango ripens.



3.Haden

Haden fruits are sweet and sour with a mildly bitter aftertaste. Pulp of the fruit is firm with a fruit-like aroma and medium fibers. Haden is mainly cultivated in Mexico, Peru, and Ecuador. These bright red mangoes are oval to round in shape and have green-yellow shaded skin with small white dots.

4. Keitt

Sweet and fruity in flavour with firm juicy flesh having citrusy aroma and little fibre. Mexico, Ecuador, the United States, and Brazil are primary sources of Keitt mango. You can relish the fruit from March to April, August, and September. They are popular in Asian cuisines, where they are enjoyed in pickles.



5. Kent

Fruits have a combination of sweet and sour flavours. The flesh is tender, juicy and has a tropical peachy aroma with limited fibre.

6. Tommy Atkins

These mangoes have tart flavour with sweet notes. The firm fibrous flesh of these fruits has a unique citrus fragrance. Tommy Atkins mango is native to Florida. It is the most extensively grown variety in the United States. This fruit is primarily grown in Brazil, Mexico, Guatemala, and Ecuador. You can have this delicious mango variety from March to July.



7. Carabao

Sweet fruits with juicy soft flesh and less fiber. Carabao mango belongs to the Philippines. It is also known as Manila mango, champagne, or Philippine mango. It is the national fruit of the Philippines, and the country ranks number six after India, China, Pakistan, Brazil, and Mexico in its production.

Varieties of Mangoes cultivated in India:

India is famed for its exclusive varieties, exotic mango hotspots, promising delightful flavours and aroma. Mango-lovers are always eager to explore and enjoy exquisite varieties of the king of fruit.

Bombay Green Mangoes – Punjab

Sadda Punjab is another hotspot for mango-lovers, famous for its Bombay Green Mangoes. These medium-sized green mangoes, available from May to July, never fail to attract their lovers from distant places.



Fazli Mangoes – Bihar/West Bengal

Known for being relatively larger than the other mangoes with lots of pulp, the right amount of sweetness and each of them going up more than a kilo, Fazli Mangoes are mostly grown in the city of Malda in West Bengal.



Badami Mangoes – North Karnataka

The northern part of Karnataka is famous for its Badami variety of mangoes. Their delicious taste can be stated from the fact that Badami is also called the Karnataka-Alphonso in the neighboring areas. They are usually available from May to July.



Paheri/Pairi Mangoes – Gujarat

Endorsed by local shopkeepers in Gujarat as pure honey, Pairi Mangoes are usually available early in the mango season during April and July. Although not as sweet as honey, a bite of this mango is very distinct with its noticeable sourness and sweetness altogether.



Himsagar and Kishan Bhog Mangoes – Murshidabad, West Bengal

One of the Nawabi cities of West Bengal, the city of Murshidabad, is famous for its vast variety of delicious mangoes. Around 230 km from Kolkata, this city is easily accessible by roadways as well as railways. Well known for types like Kishan Bhog, Himsagar, Nawabpasand and Begumpasand, Murshidabad is an important producer and exporter region of India. Himsagar mangoes are found from May to Early June.



Chausa Mangoes – Hardoi, Uttar Pradesh

Hardoi along with its adjacent regions in Uttar Pradesh is another hotspot for mango-lovers. Around 112 km from Lucknow and accessible by road network, Hardoi is famous for Chausa variety of mangoes. Available in the months of July and August, these mangoes fall under the 'sucking' category and are exclusively famous for their yellowish colour and aroma.

Rumani Mangoes – Chennai

Sold in abundance in Tamil Nadu and a few other regions in Chennai, Rumani Mangoes sport greenish-yellow skin that offers splendid flavour and juicy texture. This fruit is extremely rich in carotenoids which is a predecessor of Vitamin A and Vitamin E and hence is good for one's health. Also, apart from devouring this yummy fruit in raw form, people now also consume Rumani Mangoes in the form of smoothies, jellies, ice creams, jams, and yoghurts.



Neelam Mangoes – Andhra Pradesh

Although grown in almost all parts of India, the tastiest and most distinct variety of Neelam mangoes comes from the state of Andhra Pradesh during May and July. Neelam Mangoes have a specific sweet smell, skin bit orange-ish in colour and are usually smaller compared to other mangoes.

Kesar Mangoes – Junagadh, Gujarat

Kesar mangoes owe their name to their saffron appearance and heavenly taste. This variety, highly renowned for their distinct sweet flavour, is considered to be the 'Queen of Mangoes'. Girnar Hills of Junagadh, Gujarat are famous for their Kesar mangoes. Situated at 320 km from Ahmedabad, these hills are accessible by road and rail network. Available from May to July, Kesar mangoes are sought after as an ingredient for exotic dishes.



Langra Mangoes – Varanasi, Uttar Pradesh

Famous for its Langra variety of mangoes, Varanasi is a renowned hotspot for mango-lovers in India. Natives narrate the tale of a farmer who cultivated this variety of mangoes. Banarasi Langra mangoes, available in June-July are known for their lemon-yellow skin tone and equally delicious taste.

Totapuri Mangoes – Bangalore, Karnataka

The electronic city of India, Bangalore, is famous for its Totapuri mangoes, also known as Bangalora or Sandersha mangoes. The city's medium-sized greenish yellow mangoes are a mango-lovers delight! Available from May to July, Totapuri is also one of the significant varieties of mangoes in our country with a distinct taste and aroma!



Safeda Mangoes – Andhra Pradesh

Safeda or Banganpali or Benishan Mango is a popular fruit in several regions in Andhra Pradesh; especially in Banganpalle town, which is evident from its name. Often called as 'The King of Mangoes in South India', this fruit is significantly larger than other varieties of mangoes commonly sold mid-season in the market and on average weighs about 350 - 400 grams. Exhibiting meaty texture, the thin and firm skin of this mango is sweet in taste and lacks fibre. Moreover, Safeda Mango is known to be rich in Vitamin A and C so it is good for health.

Raspuri Mangoes – Karnataka

Predominantly cultivated in Karnataka across Bangalore, Kolar, Srinivasapura, Ramanagara during May and June, Raspuri Mangoes are an extremely popular variety of mangoes. Distinguished by its oval shape and its reddish green skin which is almost inedible, the Raspuri Mango can beat any other mango in terms of taste and juice per fruit if cultivated and harvested at the right time.



Malgoa/Mulgoba Mangoes – Salem, Tamil Nadu

These mangoes are known for their almost round shape and surprisingly thick skin. Mostly grown around parts of Tamil Nadu especially Salem, in its peak season of July and August, this fruit is big and rich in pulp and juice. It is generally considered to be one of the best mango varieties in the world.

Lakshmanbhog Mangoes – Malda, West Bengal

Exclusively grown in the Malda district of West Bengal, Lakshmanbhog mangoes are usually available in the months of June and July. With a gleaming golden reddish skin and the right amount of sweetness, these mangoes were individually chosen by the West Bengal government to be exported to the United States.



Alphonso Mangoes – Ratnagiri, Maharashtra

Available during mid-July, Alphonso Mangoes are internationally loved for their sunshine yellow appearance and delightful taste. Named after Afonso de Albuquerque, this voluptuous king of the mangoes is one of the most consumed varieties of mangoes in India among mango lovers. Ratnagiri and its neighbouring regions Maharashtra are known to be one of the finest and exclusive hotspots for mango-lovers, widely known for their Alphonso mangoes.

Amrapali Mangoes – All Over India

A hybrid variety created in 1971, by cross breeding between Dasherri and Neelam Mangoes, Amrapali is grown in farms and orchards across India and is known for its deep red flesh once fully ripe but has a relatively short shelf life compared to other commercial variety of mangoes.



Imam Pasand Mangoes – Andhra Pradesh/Telangana/ Tamil Nadu

Grown in parts of Telangana, Andhra Pradesh and Tamil Nadu and available in the months of May and June, Imam Pasand is considered to be more of an exclusive variety and known as the King of Mangoes for its unique taste. With soft skin, distinct after-taste and an incredible flavour, Imam Pasand is truly the King of Mangoes.



Mankurad Mangoes – Goa

The most popular among mango varieties in Goa, Mankurad is a mid-season fruit with very less fibre, naturally perfectly sweetened and known among Mango connoisseurs to be better than the all-famous Alphonso Mangoes. Usually, these mangoes are available in the local markets as early as April.



Dasherri Mangoes – Lucknow & Malihabad, Uttar Pradesh

The land of Nawabs is equally popular for its royal mangoes. Lucknow, along with its adjoining towns, is widely known for its mango belt in North India. Available from mid-May to late August, Dasherri mangoes grown here are easily distinguishable from other varieties of mangoes in India by their green peels and wholesome taste.



Cultivars of mangoes in India

Andhra Pradesh	Banganapalli, Suvarnarekha, Neelam and Totapuri
Bihar	Bombay Green, Chausa, Dashehari, Fazli, Gulabkhas, Kishen Bhog, Himsagar, Zardalu & Langra
Gujarat	Kesar, Alphonso, Rajapuri, Jamadar, Totapuri, Neelam, Dashehari & Langra
Haryana	Chausa, Dashehari, Langra and Fazli
Himachal Pradesh	Chausa, Dashehari and Langra
Karnataka	Alphonso, Totapuri, Banganapalli, Pairi, Neelam and Mulgoa
Madhya Pradesh	Alphonso, Bombay Green, Dashehari, Fazli, Langra and Neelam
Maharashtra	Alphonso, Kesar and Pairi
Punjab	Chausa, Dashehari and Malda
Rajasthan	Bombay Green, Chausa, Dashehari and Langra
Tamil Nadu	Alphonso, Totapuri, Banganapalli and Neelam
Uttar Pradesh	Bombay Green, Chausa, Dashehari and Langra
West Bengal	Fazli, Gulabkhas, Himsagar, Kishenbhog, Langra & Bombay Green

Varieties of Mangoes grown in Karnataka

Srinivaspur in Kolar (Karnataka) is nicknamed as the Mango City of India as over 63 species of mangoes are found here. Banganapalli, Mallika, Alphonso/Badami, Totapuri, Neelam, Rasputri, Malgova, Amlet (pickle variety) are famous mango varieties found in Karnataka.

More information on Mango cultivars grown in the world and India is available on National Mango database <https://mangifera.res.in/varieties.php>. There are about 12 mango varieties which have been GI tagged for particular states of India. Info:

https://apeda.gov.in/apedawebsite/about_apeda/Mango_brochure.pdf

Top Mango producing countries of the world 2020:

Rank	Country	Share in Export value	Export Value (USD Mn)
1	India	45.22%	24.75 M
2	Indonesia	6.61%	3.62 M
3	Mexico	4.34%	2.37 M
4	China	4.33%	2.37 M
5	Pakistan	4.28%	2.34 M
6	Brazil	3.9%	2.14 M
7	Malawi	3.54%	1.94 M
8	Thailand	3.03%	1.66 M
9	Bangladesh	2.65%	1.45 M
10	Egypt	2.55%	1.40 M

State wise Production of Mangoes (2019-20)

Rank	State	Production Quantity
1	Uttar Pradesh	40 Lakh tonnes
2	Andhra Pradesh	5001.74 MT
3	Karnataka	8 Lakh tonnes
4	Bihar	2434.65 MT
5	Gujarat	1219.86 MT
6	Tamil Nadu	1250.77 MT
7	Odisha	805.77 MT
8	West Bengal	844.88 MT
9	Jharkhand	432.93 MT
10	Maharashtra	566.37 MT

Source: <https://theindianblog.in/top-10-states-india/top-10-mango-producing-states-in-india/>

Top 10 exporting countries (2020)

Rank	Country	Share in Export value	Export Value (USD Mn)
1	Thailand	15.8%	570.70
2	Mexico	12.55%	453.24
3	Netherlands	11.37%	410.67
4	Peru	8.21%	296.44
5	Brazil	7.06%	254.97
6	Vietnam	5.1%	184.06
7	Hong Kong	3.98%	143.62
8	India	3.82%	138.04
9	Spain	3.12%	112.53
10	Pakistan	2.81%	101.46

India is also a prominent exporter of fresh mangoes to the world. The country has exported 21,033.58 MT of fresh mangoes to the world valued at Rs. 271.84 crores/36.23 USD Millions during the year 2020-21.

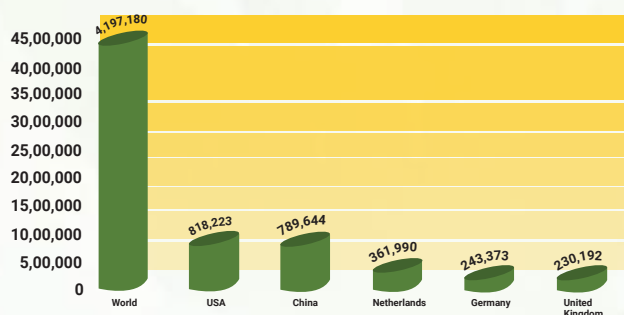
Top importing countries of Mango from India (2020-21)

Rank			
1	U Arab Emts	12,923.69	14,472.70
2	U K	3,376.89	7,337.83
3	Qatar	2,116.64	2,332.75
4	Oman	1,693.15	1,898.19
5	Kuwait	773.27	1,747.02
6	Nepal	4,659.48	952.55
7	Singapore	350.24	718.24
8	Canada	295.96	541.58
9	Baharain Is	342.02	398.77
10	Saudi Arabia	337.37	376.56
	Page Total	26,868.71	30,776.19

080450: Fresh or dried guavas, mangoes, and mangosteens.

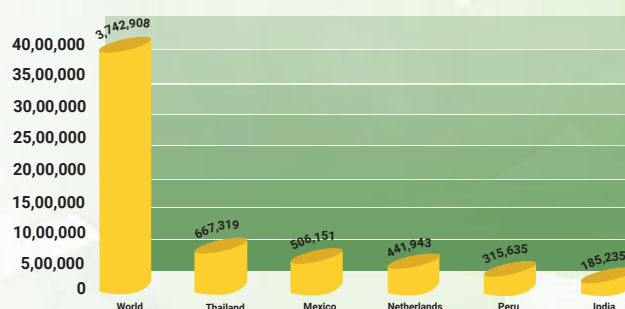
2021 Global Imports 4,197,180 USD.

No	Country	Value (USD)
1	USA	818,223
2	China	789,644
3	Netherlands	361,990
4	Germany	243,373
5	United Kingdom	230,192



2021 Global Exports 3,742,908 USD.

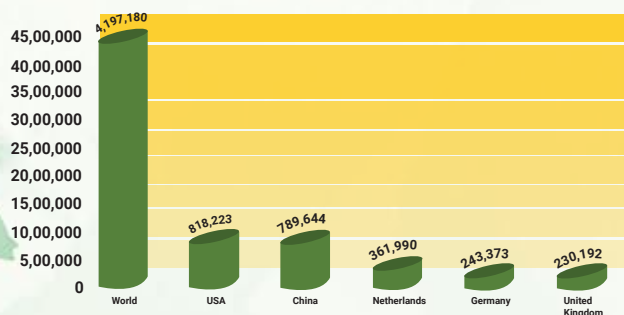
No	Country	Value (USD)
1	Thailand	667,319
2	Mexico	506,151
3	Netherlands	441,943
4	Peru	315,635
6	India	185,236



081290: Fruit and nuts, provisionally preserved, e.g., by sulphur dioxide gas, in brine, in sulphur water or in other preservative solutions, but unsuitable in that state for immediate consumption (excluding cherries).

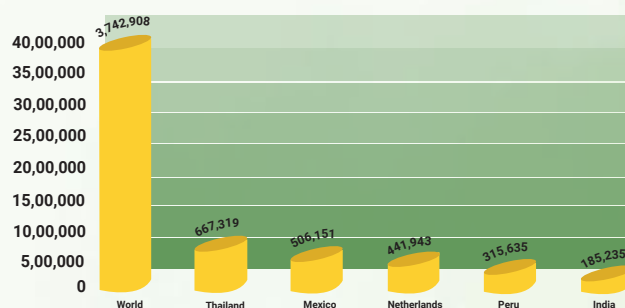
2021 Global Imports 115,170 USD.

No	Country	Value (USD)
1	Japan	26,175
2	China	16,586
3	Austria	12,293
4	United Kingdom	10,569
5	Germany	10,026



2021 Global Exports 139,671 USD.

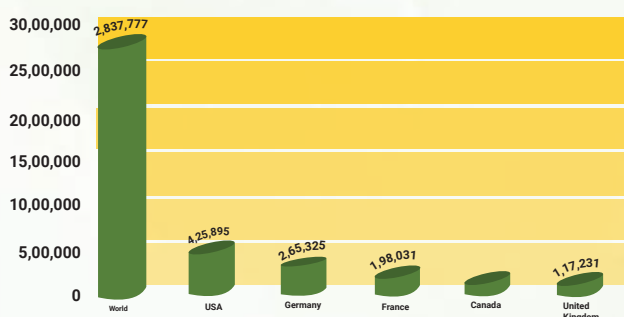
No	Country	Value (USD)
1	China	41,614
2	Spain	25,706
3	Dominican Republic	15,515
4	Vietnam	13,723
23	India	520



200799: Jams, jellies, marmalades, purées or pastes of fruit, obtained by cooking, whether or not containing added sugar or other sweetening matter(excluding citrus fruit and homogenized preparations of subheading 2007.10)

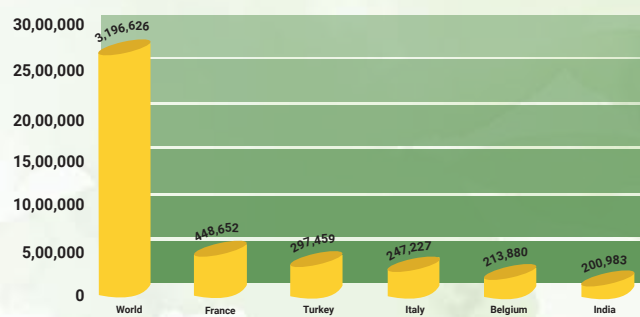
2021 Global Imports 2,837,777 USD.

No	Country	Value (USD)
1	USA	425,895
2	Germany	265,325
3	France	198,031
4	Canada	148,853
5	United Kingdom	117,231



2021 Global Exports 3,196,626 USD.

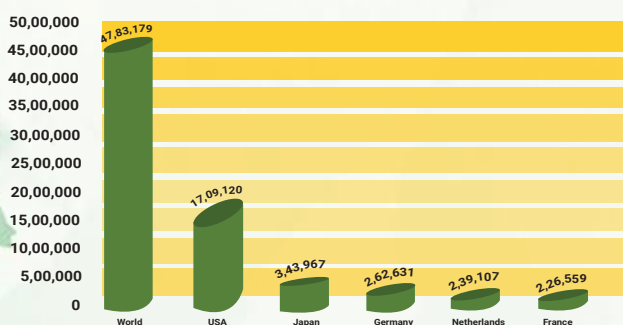
No	Country	Value (USD)
1	France	448,652
2	Turkey	297,459
3	Italy	247,227
4	Belgium	213,880
5	India	200,983



200899 Fruit and other edible parts of plants, prepared or preserved, whether or not containing added sugar or other sweetening matter or spirit (excluding prepared or preserved with vinegar, preserved with sugar but not laid in syrup, jams,fruit jellies, marmalades, fruit purée and pastes, obtained by cooking, and nuts, groundnuts and other seeds, pineapples, citrus fruits, pears, apricots, cherries, peaches, strawberries, palm hearts and cranberries)

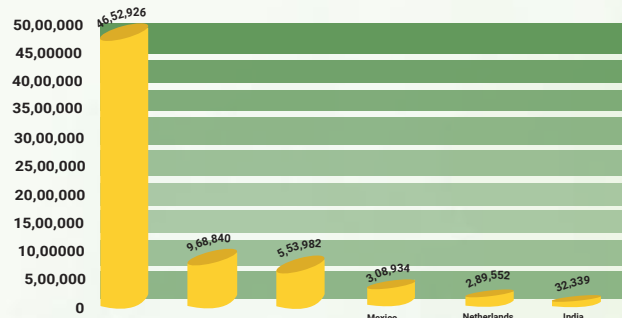
2021 Global Imports 4,783,179 USD.

No	Country	Value (USD)
1	USA	1,709,120
2	Japan	343,967
3	Netherlands	262,631
4	Germany	239,107
5	France	226,559



2021 Global Exports 4,652,926 USD.

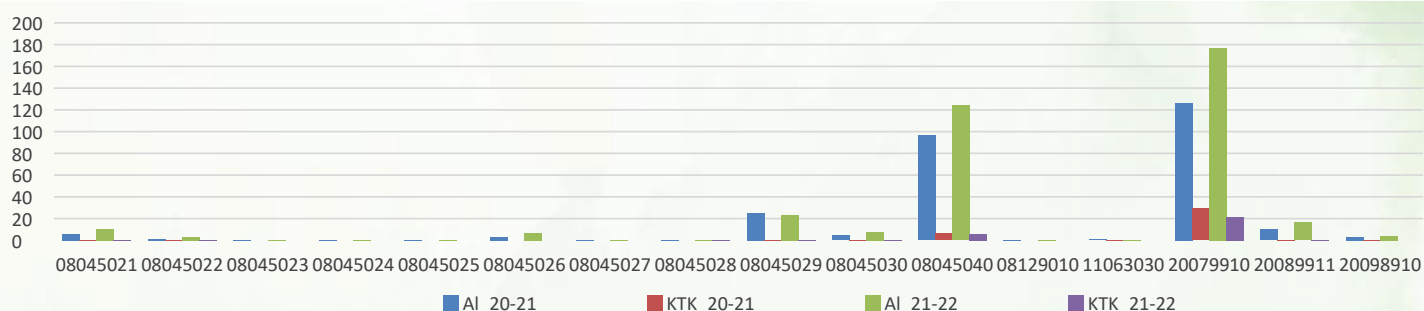
No	Country	Value (USD)
1	China	968,840
2	Korea	553,982
3	Mexico	308,934
4	Netherlands	289,552
27	India	32,339



Export performance of Mango 2020 to 2022: All India and Karnataka.

No.	HS Code	Description	2020-21		2021-22		Major Importing Countries
			AI	KTK	AI	KTK	
1	08045021	Alphonso (Hapus)	6.08	0.02	10.12	0.09	USA, UAE & Kuwait
2	08045022	Baganapalli	1.46	0.04	3.02	0.18	UAE, UK and Oman
3	08045023	Chausa	0.05	Nil	0.05	Nil	Japan, UAE, UK
4	08045024	Dasheri	0.09	Nil	0.11	Nil	UAE, UK and Qatar
5	08045025	Langda	0.08	Nil	0.16	Nil	UAE, UK & Japan
6	08045026	Kesar	2.92	Nil	6.93	Nil	UK, Canada & UAE
7	08045027	Totapuri	0.07	0.00	0.17	0.00	UK, Qatar & UAE
8	08045028	Mallika	0.04	0.00	0.09	0.01	UAE, Japan & Kuwait
9	08045029	Other Mangoes	25.42	0.07	23.55	0.02	UAE, Arab & UK
10	08045030	Mangoes, Sliced dried	4.52	0.17	7.66	0.36	USA, Nepal & UAE
11	08045040	Mango Pulp	96.43	6.21	124.01	5.54	Saudi, Netherland & Yemen Rep
12	08129010	Mango slices in Brine	0.29	0.00	0.48	Nil	UK, Netherland & USA
13	11063030	Flour of Mango	0.83	0.01	0.61	0.00	USA, Canada & UK
14	20079910	Jams, Jellies, Marmalade etc. of Mangoes	126.21	29.21	177.04	21.74	Netherland, Saudi & UAE
15	20089911	Mango Squash	9.98	0.04	16.32	0.03	UAE, Nepal & Netherlands
16	20098910	Mango Juice	3.07	0.01	4.20	Nil	Netherland, USA & Korea

* Value in Mn USD.



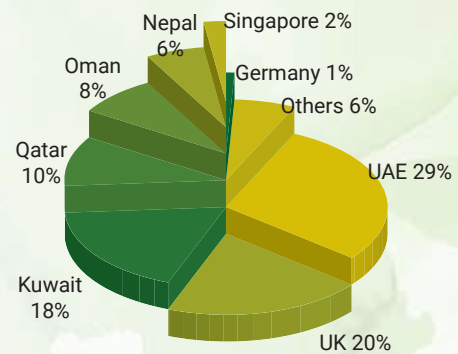
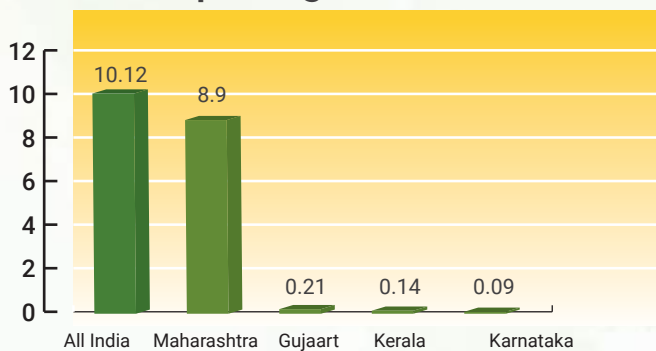
Source: Exim Analytics, DGCIS.

Export Analysis of Mangoes: 2021-22

08045021: Alphonso (Hapus).

No.	State	Value USD Mn	Destination
0	All India	10.12	USA, UAE, Kuwait (+28)
1	Maharashtra	8.90	UAE, UK, Kuwait (+22)
2	Gujarat	0.21	UK, Canada, Netherlands (+16)
3	Kerala	0.14	UAE, Qatar, Singapore (+3)
4	Karnataka	0.09	Saudi Arabia, UK, Qatar

Exporting States

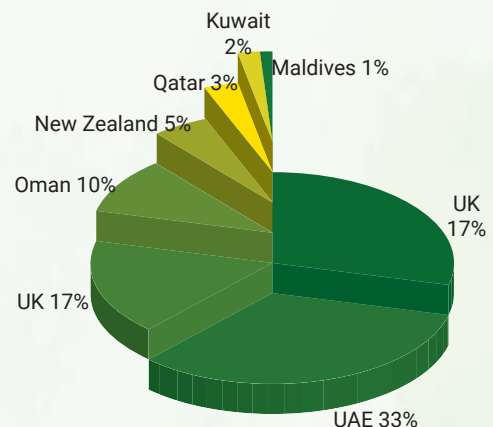
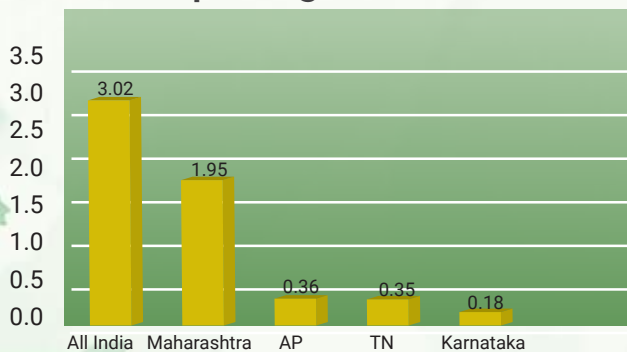


Importing Countries

08045022: Baganapalli

No.	State	Value USD Mn	Destination
0	All India	3.02	UAE, UK, Oman (+21)
1	Maharashtra	1.95	UAE, Oman, UK (+17)
2	Andhra Pradesh	0.36	New Zealand, Germany, Canada (+6)
3	Tamil Nadu	0.35	Maldives, Singapore, Malaysia (+6)
4	Karnataka	0.18	UK, Australia, Germany (+7)

Exporting States

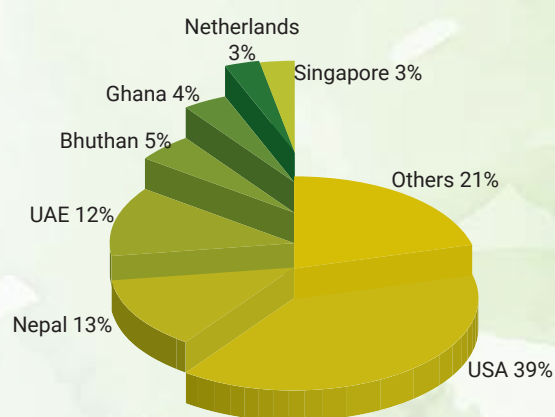
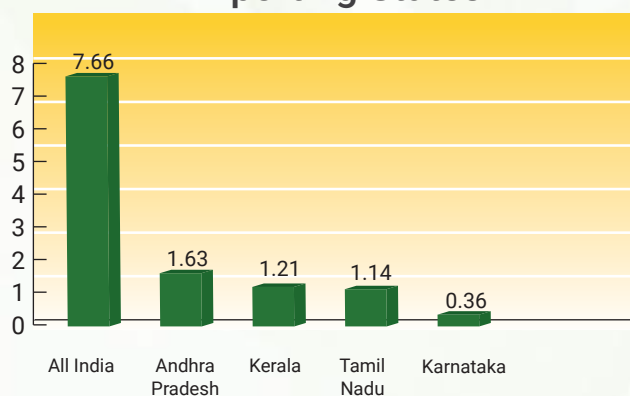


Importing Countries

08045030: Mangoes, Sliced dried.

No.	State	Value USD Mn	Destination
0	All India	7.66	USA, Nepal, UAE (+36)
1	Andhra Pradesh	1.63	USA, Netherlands, Germany (+7)
2	Kerala	1.21	UAE. Qatar, Kuwait (+14)
3	Tamil Nadu	1.14	USA, Singapore, UAE (+9)
7	Karnataka	0.36	Poland, Qatar, UK (+9)

Exporting States

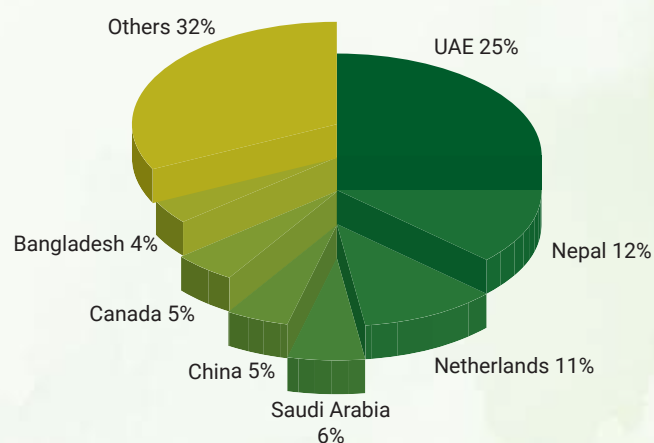
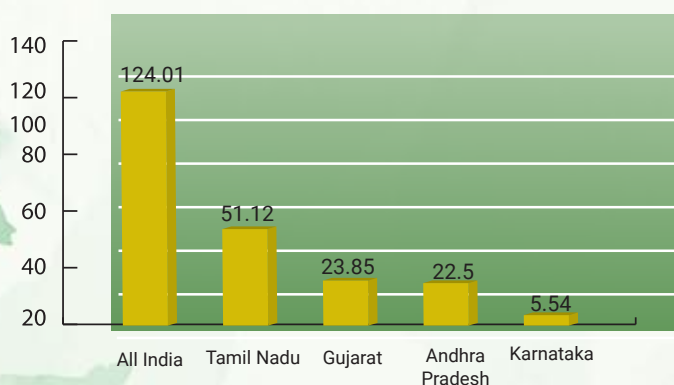


Importing Countries

08045040: Mango Pulp.

No.	State	Value USD Mn	Destination
0	All India	124.01	Saudi, Netherland, Yemen Rep (+88)
1	Tamil Nadu	60.74	Netherlands, UAE, Russia (+65)
2	Gujarat	48.87	Netherlands, Saudi Arabia, China (+68)
3	Andhra Pradesh	32.17	Netherlands, Saudi Arabia, USA (+40)
5	Karnataka	21.74	Netherlands, USA, France (+30)

Exporting States

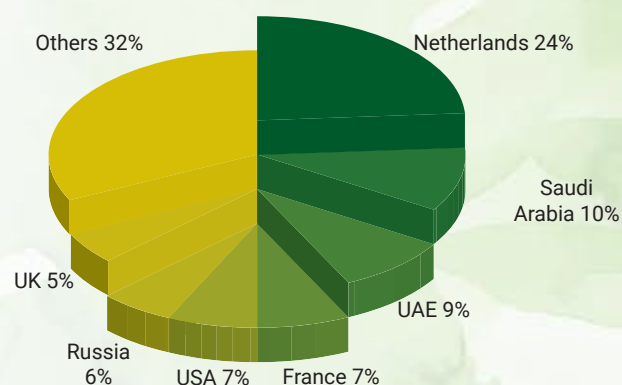
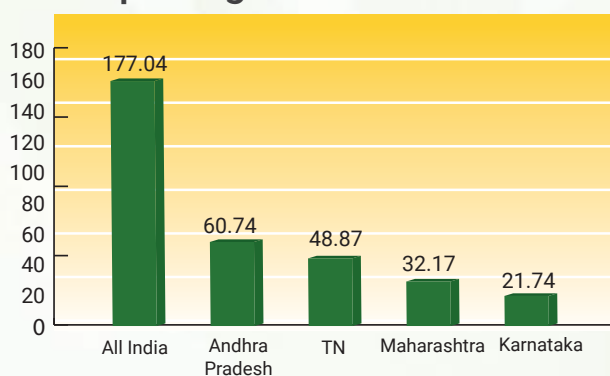


Importing Countries

20079910: Jams, Jellies, Marmalade etc. of Mangoes

No.	State	Value USD Mn	Destination
0	All India	177.04	Netherlands, Saudi Arabia, UAE (+83)
1	Andhra Pradesh	60.74	Netherlands, UAE, Russia (+65)
2	Tamil Nadu	48.87	Netherlands, Saudi Arabia, China (+68)
3	Maharashtra	32.17	Netherlands, Saudi Arabia, USA (+40)
4	Karnataka	21.74	Netherlands, USA, France (+30)

Exporting States

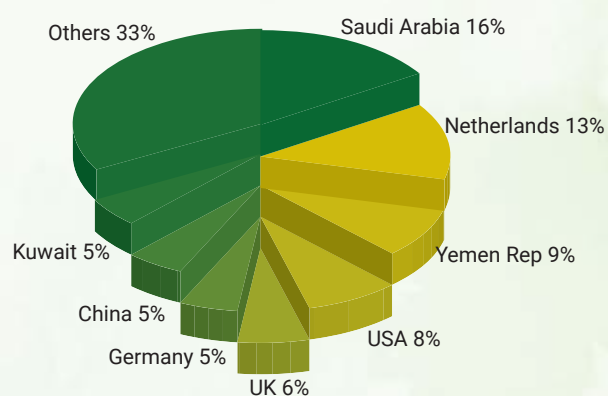
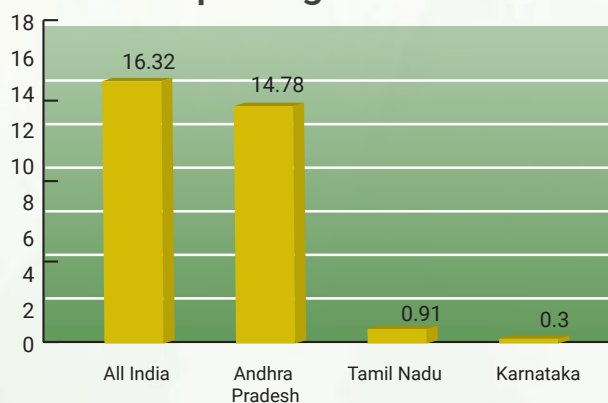


Importing Countries

20089911: Mango Squash

No.	State	Value USD Mn	Destination
0	All India	16.32	UAE, Nepal, Netherlands (+39)
1	Andhra Pradesh	14.78	UAE, Netherlands, Saudi Arabia (+21)
2	Tamil Nadu	0.91	Nepal, Liberia, Taiwan (+4)
3	Karnataka	0.30	Nepal, Bangladesh, Maldives

Exporting States



Importing Countries

SPS -TBT Standards for Mangoes

-  Guidelines for different countries - Fresh Mango Export Processes – Tridge:
<https://www.tridge.com/guides/mango/topics/export-processes>
-  Product profile of Mangoes: APEDA Agri-Exchange:
https://agriexchange.apeda.gov.in/product_profile/prd_profile.aspx?categorycode=0204
-  Export standards APEDA: Packaging and details of Mangoes
https://agriexchange.apeda.gov.in/product_profile/Packaging.aspx?categorycode=0204
-  Export standards APEDA: Packaging and details of Mangoes
https://agriexchange.apeda.gov.in/product_profile/Packaging.aspx?categorycode=0204
-  Quality standard guidelines for Mango exports to Japan:
https://apeda.gov.in/apedawebsite/Announcements/Guidelines_JAPAN_2006.pdf

General standards for Mangoes

-  Whole, firm, ripe and fresh in appearance
-  Produce affected by rotting or any other deterioration rendering it unfit for consumption is excluded
-  Clean, practically free from visible foreign matter
-  Free from black necrotic spots or traces extending below the skin
-  Free from pronounced bruising
-  Practically free from damage caused by pests infesting the fresh fruit
-  Free from damage caused by low temperatures.
-  Free from abnormal external moisture, except condensation after removal from cold storage
-  Free from any foreign smell and/or taste
-  Sufficiently developed and with a satisfactory degree of ripeness.

Source: Researchgate publications

https://www.researchgate.net/publication/356593730_UNDERSTANDING_SPS_REQUIREMENTS_FOR_GHANA'S_EXPORT_TO_THE_EU_FOCUS_ON_MANGO_PRODUCTS

Processing technologies available

-  Carbonated fruit beverages from selected fruits (mango, grapes, lime, orange):
<https://cftri.res.in/technologies/FVP/cbj Juice.pdf>
-  CFortified Mango bar: <https://cftri.res.in/technologies/FVP/fmb.pdf>
-  Alphonso Mango - export by Ship: Ship: <https://cftri.res.in/technologies/FVP/am.pdf>
-  Alphonso Mango - export by Ship:
[https://uphorticulture.in/DPR/Grain%20processing/Mango%20Pulp%20\(Fruit%20Pulp\).pdf](https://uphorticulture.in/DPR/Grain%20processing/Mango%20Pulp%20(Fruit%20Pulp).pdf)
-  Mango Jam:
Jam: <https://discoverfoodtech.com/jam-production-process-with-flow-chart/>
-  Mango squash, Pickle, Amchur, Powder, Mango candy, Spicy mango bar etc.from TNAU:
https://agritech.tnau.ac.in/postharvest/pht_fruits_mango.html
-  Handbook of preparation of Mango Ready to Serve Drink (RTS) from IIFPT:
<http://www.iifpt.edu.in/pmfme/mangocurmet.pdf>



Measures for Enhancing Competitiveness for Exports

-  India's exports of fresh mangoes increased from 20,302 tonnes in 1987-88 to 46,789.60 tonnes in 2019-20.
-  India grows almost 1,000 mango varieties, but uses only 30 commercially, including the famous Alphonso.
-  The value of India's mango exports slumped from \$56 million in 2019-20 to \$28.3 million in April-Feb 2020-21.
-  According to a report by the National Horticulture Board, Mexico, Haiti and Brazil account for the majority of North America's imports. India and Pakistan are the predominant suppliers to West Asia. Southeast Asian countries get their supplies mostly from the Philippines and Thailand. European Union sources mangoes from South America and Asia.
-  APEDA, an autonomous organization under the Department of Commerce, has been mandated with the export promotion of mangoes. It has taken several measures to push India's exports. According to APEDA's website, Agri Export Zones have been established in almost all mango growing areas and packhouses on modern lines have been provided in all mango exporting regions, including in Ratnagiri and Sindhudurg in Maharashtra and in Navsari and Borsad in Gujarat for Alphonso variety; in Latur and Aurangabad for the Kesar mango; in Saharanpur and Malihabad in Uttar Pradesh for Dashehari and Chausa mangoes.
-  Also, facilities for facilitating mango exports like a Post-harvest Management Centre have been established at Malihabad and Saharanpur. Similarly, a mango Export Facility Centre has been established at Ratnagiri and mango farmers of Alphonso and Kesar have been trained in global standards requirements.
-  While there is no shortage of availability of refrigerated containers for sea shipments, mangoes are exported by air to distant destinations like the EU and USA. To facilitate exports of mangoes by refrigerated vans, APEDA provides financial assistance for purchase of insulated/refrigerated transport/mobile pre-cooling units. Assistance for Reefer transport vehicles is also available under the Mission for Integrated Development of Horticulture (MIDH), a centrally sponsored scheme implemented by the Ministry of Agriculture & Farmers Welfare.

- ✿ In June 2021, a week-long Indian mango promotion program was organized in Bahrain where 16 varieties of the fruit including three GI certified Khirsapati & Lakshmanbhog (West Bengal), Zardalu (Bihar) were displayed. The varieties of mangoes were sold through 13 stores of the group in Bahrain. The mangoes were sourced from farmers in Bengal and Bihar by APEDA registered exporter.
- ✿ Since Covid-19 pandemic had hit physical export promotion programs, APEDA was organizing virtual meets to provide a platform to the exporters and importers of mangoes from India during 2020-21.
- ✿ APEDA organized Mango festival in Berlin, Germany.
- ✿ In a bid to increase mango exports to South Korea, APEDA in collaboration with Indian embassy, Seoul and Indian Chamber of Commerce in Korea, had earlier organized a Virtual Buyer Seller Meet.
- ✿ For the first time in this season, India has shipped a consignment of 2.5 Metric Tonnes (MTs) of GI certified Banganapalli & other variety Survarnakha mangoes sourced from farmers in Krishna & Chittoor districts of Andhra Pradesh to South Korea.
- ✿ Mango in India is also referred to as 'king of fruits' and referred to as Kalpavriksha (wish granting tree) in ancient scriptures. While most of the state in India have mango plantations, Uttar Pradesh, Bihar, Andhra Pradesh, Telangana, Karnataka account for a major share in total production of the fruit.
- ✿ Another way to push exports is innovation in food processing. This will increase the shelf life of the fruit and also help in overcoming the issue of distance. Alphonso, Kesar, Totapuri and Banganapalli are leading export varieties from India. Products like mango juice/squash, Jam, powder, amchur, mango pulp, and mango slice etc., can also be exported.

Source: Ministry of Commerce & Industry:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1735899>



Responsibility Matrix

Sl.no	Actions	Responsibility
1	Research on best global varieties	APEDA India
2	Delegation of progressive farmers to Pakistan, South America and Philippines.	Horticulture department, Karnataka and Mango Board.
3	Farmer Sensitization program on available opportunities	APEDA, India and Mango Board, Karnataka.
4	Inviting Exporters/entrepreneurs to Investors meet for contract farming and processing industries	APEDA, India and Mango Board, Karnataka.
5	Farmer database and aggregation of the produce in Karnataka	Mango Board, Karnataka, Horticulture department and related FPOs
6	Processing technology and Value Addition Handholding.	Mango Board through APEDA India, CFTRI, Mysore and IIFPT, Thanjavur
7	Export training, orientation, and Market Intelligence	APEDA and VTPC Karnataka

Regulators and Service providers

No.	Organization	Service	Contact
1.	APEDA	RCMC, Market survey and assistance.	1st Floor, Beeja Bhavan, Bellary Rd, Hebbal, Bengaluru - 560024. E: apedabl@apeda.gov.in
2.	KAPPEC	PMFME Scheme	17, Richmond Rd, Shanthala Nagar, Richmond Town, Bengaluru - 560025. E: kapppec1996@gmail.com
3.	DGFT	IEC, Customs and ICEGATE	6th floor, Kendriya Sadan, C & E Wing, 17th main, Koramangala 2nd Block, Koramangala, Bengaluru - 560034 E: bangalore-dgft@nic.in
4.	Plant Quarantine	Phyto Sanitary certification (SPS)	Hebbal-Boopasandra Road HA Farm Post, Bengaluru - 560024. E: dd-pqfsb-ka@nic.in
5.	BIAL Cool Port	Freight and Flight	KIAL Road, Devanahalli, Bengaluru -560300 W: http://www.aisats.in
			Karnataka State Mango Development and Marketing Corporation Ltd. F, Block, 2nd floor, Cauvery Bhavan, K.G. Road, Bengaluru- 560009 E: ksmdmcl.helpdesk@ka.gov.in